MSE 311 Thermodynamics of materials-Homework 9

Name: *Student ID:*

1. Carbon has the following three allotropes: graphite, diamond, and a metallic form called solid III. Graphite is the stable form of 1000 K and 1 atm pressure, and increasing the pressure on graphite at temperatures less than 1440 K causes the transformation of graphite to diamond and then the transformation of diamond to solid III. Calculate the pressure which, when applied to one mole graphite at 1000 K, causes the transformation of graphite to diamond, given

$H_{1000K,graphite} - H_{1000K,diamond} = -1896J$

$S_{1000K,graphite} = 5.74J/K$

$S_{1000K,diamond} = 2.36J/K$

Density of graphite at 1000K: 2.62 g/cm^3

Density of diamond at 1000K:3.52 g/cm^3

2. Derive the following formula， considering Maxwell equation and van der Waals Gas.

$$\Delta H_{evap} = -a\left(\frac{1}{v_v} - \frac{1}{v_l}\right) + P(v_v - v_l)$$

3. Show that the truncated Kamerlingh Onnes virial equation

$$\frac{PV}{RT} = 1 + \frac{B'(T)}{V}$$

Reduces to *P(V-b')=RT*, and find the expression for *b'*.

4. Rewrite the van der Waals equation into the following form of virtual equation and gives out the parameter B.

$$\frac{PV}{RT} = 1 + B \cdot P + C \cdot P^2 + D \cdot P^3 + \ldots$$

5. Assuming that nitrogen behaves as a van der Waals gas with a = 1.391 (liters)^2·atm/mole^2 and b = 39.1 cm^3/mole, calculate the change in the Gibbs free energy when the volume of 1 mole of nitrogen is increased from 1 to 2 liters at 400 K. If the nitrogen had behaved as an ideal gas, what is the changes in Gibbs free energy?

6. Derive the following formula from Gibbs-Duhem equation.

$$\Delta G^M = RT\left[X_A \ln X_A + (1 - X_A)\ln(1 - X_A)\right]$$

7. Using the virial equation of state for hydrogen at 298 K is $PV=RT(1 + 6.4\times10^{-4}P)$, calculate

a. The fugacity of hydrogen at 500 atm and 298 K,

b. The pressure at which the fugacity is twice the pressure,

c. The change in the Gibbs free energy caused by a compression of 1 mole of hydrogen at 298 K from 1 to 500 atm

d. What is the magnitude of the contribution to (c) caused by the nonideality of hydrogen?